Supersedes
16A47(Aer)
Dated 15 March 1948

MILITARY SPECIFICATION
ATTENUATOR, CN-45(*)/UP

This specification has been approved by the Bureau of Aeronautics, Department of the Navy.

1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment covered by this specification shall be Attenuator, CN-45(*)/UP, for use in maintenance and testing AN/APS-20 radar units. Specifically, the attenuator shall have two uses:

(1) As a resistance load for the MD-53/APS-20 modulator.

(2) As a magnetron conditioner on the AN/APS-20 system.

* The parenthesis in the type designation shall be replaced by a letter furnished by the procuring agency upon application by the contractor in accordance with the requirements of paragraph 3.2.9 after each contract has been awarded. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

1.2 Classification - Attenuator CN-45(*)/UP shall be of one type, consisting of the instrument itself and accessories as follows:

Quan Per Equipment	Nomenclature	Type Designation	Max Allow. Size (in.)	Max Allow. Wt (lb)	Applicable Paragraph
1	Attenuator	CN-45(*)/UP	18 W, 9-1/2 D, 19 H	32 including all accessories	3.3.5
1	Cable, Pulse	-	36	-	3.3.6
1	Cable, Octopus	-	20	-	3.3.7

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 General - The following specifications, standards, and other publications of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-T-945	Test Equipment, For Use With Electronic Equipment; General Specification
JAN-C-17	Cables, Coaxial and Twin Conductor for Radio Frequency
JAN-P-658	Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical, and Electromechanical)

Bureau of Aeronautics

SR-197	Nameplates and Nomenclature; Aircraft Electronic Equipment

OTHER PUBLICATIONS

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Bureau of Ships

RE49F353	Receptacle, Pulse, UG-37A/U Type B, Air-to-Air

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RE49F354	Plugs, High Voltage, UG-36/U for Use With Cable RG-27/U; UG-174/U for Use With Cable RG-28/U

Air Force-Navy Aeronautical

AN3102	Connectors - Electrical, Receptacle, Box Mounting
AN3106	Connectors - Electrical, Plugs, Straight

2.2 Availability of Specifications, Other Publications and Drawings

(1) When requesting specifications, other publications, and drawings, refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification, and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Air Station, Johnsville, Pennsylvania, Attn: Technical Records Division.

(3) Copies of Bureau of Ships drawings may be obtained upon application to Bureau of Ships, Department of the Navy, Washington 25, D. C., Attn: Electronics Division.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviations from this specification or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials

3.1.1 General - The equipment shall fulfill the following requirements for parts and materials:

(1) Design Objectives - Minimum size and weight, simplicity of operation, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials, parts, and processes other than those required by Specification MIL-T-945 shall be investigated, and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, or reliability can be realized by their substitution, a request for approval of a nonstandard part shall be submitted to the procuring agency for consideration. Each request shall be accompanied by complete supporting information.

(2) Selection of Parts and Materials - The fulfillment of major design objectives shall govern the selection of parts and materials. All parts and materials shall meet the requirements of Specification MIL-T-945; however, if the contractor can demonstrate that the use of such parts and materials will not fulfill the major design objectives, then parts and materials shall be used which most nearly meet or exceed the requirements of Specification MIL-T-945.

(3) Approval for Nonstandard Parts and Materials - Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-T-945. Requests for approval must be forwarded to the procuring agency by the contractor before parts are purchased or fabricated.

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(4) Workmanship - Workmanship, fabrication, and manufacturing process shall conform to the requirements of Specification MIL-T-945.

3.2 General Design Requirements -

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-T-945 with the following exceptions:

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions, or reasonable combinations of service conditions, as specified in MIL-T-945 except that the requirements specified shall apply only within the temperature range from -40° to +55° C.

3.2.3 Operating Power - No external input power source is required.

3.2.4 Operating Life -

3.2.4.1 Reliable Operating Life - The equipment shall be capable of operation, as specified herein for a period of 200 hours without requiring servicing or replacement of parts. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.4.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing or replacement of parts. The life of such parts and parts requiring replacement within this interval shall be specified by the manufacturer.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for at least 200 hours, continuously or intermittently, without necessity for adjustment of controls other than those located on the front panel.

3.2.6 Cables and Connectors - Cables and connectors shall be in accordance with the requirements of Specification MIL-T-945.

3.2.7 Fungus and Moisture Treatment - The equipment shall be moisture and fungus-proofed by selection of parts and materials that are non-nutrient for fungus. The parts and materials shall be treated so that prior to their use in the equipment, over-all spraying of the equipment is not necessary.

3.2.8 Interchangeability - Mechanical and electrical interchangeability shall exist between similar assemblies, subassemblies, and replaceable parts regardless of the manufacturer or supplier. Interchangeability for the purposes of this paragraph does not mean identity, but requires that a substitution of such like assemblies, subassemblies, and replaceable parts be easily effected without physical or electrical modifications to any part of the equipment including cabling, wiring, and mounting. In the design of the equipment, provisions shall be made for design tolerance to accommodate various articles such as tubes, resistors, and other components having the limiting dimensions and characteristics set forth in the specifications for the particular component involved without departure from the specified performance.

3.2.9 Nameplates and Nomenclature - Nameplates and nomenclature shall be in accordance with the requirements of Specification SR-197..

3.3 Detail Design Requirements

3.3.1 General - Attenuator CN-45(*)/UP shall consist of a bank of resistors and an inductance, mounted in a frame, suitably protected from injury by a metal grill covering, and provided with suitable connectors. This equipment shall be capable of being used as either a modulator load or as an attenuator and shall fulfill the following detail design requirements:

3.3.2 Performance - When used as a modulator load, the unit shall present a noninductive resistance of 25 ohms ±10 percent to the modulator. When used as an attenuator, the unit shall be connected to form a slightly inductive impedance of 10 ohms ±10 percent.

3.3.3 Power Dissipation - When used as a modulator load, the unit shall be capable of dissipating 1200 watts (average power) continuously without the use of forced draft ventilation. When used as an attenuator, the unit shall be capable of dissipating 500 watts (average power) continuously without the use of forced draft ventilation.

3.3.4 Insulation - The equipment shall be so insulated that a voltage pulse of 10 kilovolts can be applied to the input terminal without causing breakdown when used as a modulator load. The equipment shall be so

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insulated to withstand a voltage pulse of 10 kilovolts applied between line parts and ground when used as an attenuator.

3.3.5 Attenuator CN-45(*)/UP

3.3.5.1 Form Factor and Weight - The dimensions of the equipment shall not be greater than 18 inches wide, 9-1/2 inches deep, and 19 inches high. The over-all weight of the unit shall be no more than 32 pounds.

3.3.5.2 Contents of Unit - The unit shall contain 10 resistors and an inductance, with suitable connectors and mountings. The resistors shall be noninductive, 10-ohm, 160-watt, and ferrule-mounted. The inductance shall consist of 30 turns of No. 16 AWG enameled wire, close-wound in a single layer on a 3/4-inch diameter form.

3.3.5.3 Electrical Connections to Unit - Two UG-37A/U (Bureau of Ships Drawing RE49F353) electrical connections shall be provided, one labeled "TRANSMITTER CABLE (IF USED)" and the other "MODULATOR CABLE." Suitable protective caps shall be provided for both connectors with captive chains.

3.3.5.4 Electrical Connectors Within Unit - Conversion from a modulator load to an attenuator shall be accomplished by changing internal connections.

(1) Modulator Load - When the equipment is used as a modulator load, internal connection shall be such that the modulator cable receptacle presents a load of 25 ohms ±10 percent. Five parallel pairs of resistors shall be connected in series.

(2) Attenuator - When the equipment is used as an attenuator, internal connections shall be such that two parallel pairs of resistors shall be connected in series with the inductance to provide a resistance of approximately 10 ohms between receptacles.

3.3.5.5 Case - The equipment shall be provided with a combination case meeting the requirements of Specification MIL-T-945 except that the sides shall be of a grill construction for suitable ventilation. Large nameplates with the word "DANGER" in large red letters shall be attached to the two wide sides of the case.

3.3.6 Pulse Cable - A pulse cable, consisting of a 3-foot length of RG-28/U cable (JAN-C-17) with a UG-174/U connector at each end, shall be provided (Bureau of Ships Drawing RE49F354).

3.3.7 Octopus Cable - An octopus cable, consisting of a main lead approximately 12 inches long and a branch section approximately 8 inches long, shall be provided. One end of the main lead shall terminate in an AN3102-28-2S connector, and the other end shall terminate in an AN3106-28-2P connector. The end of the branch section shall terminate in an AN3106-14S-SP connector. Connections shall be as follows:

AN3102-28-2S Pin	Wire Size No.	AN3106-28-2P Pin
A	20	A
B	16	B
D	20	D
F	16	F
H	RG-39/U	H
M	12	M
N	16	N
P	12	P
C		C

AN3106-14S-SP

No. 20 Pin No. 20

D E

4. METHODS OF SAMPLING, INSPECTION, AND TEST

4.1 Classification - The equipment specified herein shall be subject to preproduction and inspection tests as follows:

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(1) Preproduction Tests - Preproduction tests shall be made on a sample(s) representative of the production equipments, to determine that the production equipments will meet the requirements of this specification and the applicable subsidiary specifications.

(2) Inspection Tests - Inspection tests shall be made on production models submitted for acceptance under the contract to determine conformance with the requirements of this specification and the applicable subsidiary specifications and to determine that the test sets are equivalent in performance and construction to the approved preproduction sample. Any production prior to approval of the preproduction sample shall be at the contractor's risk.

4.2 Preproduction Tests - Preproduction tests shall be conducted on two preproduction models representative of the production equipment as follows:

4.2.1 Scope of Preproduction Tests - Preproduction tests shall be conducted at a laboratory designated by the procuring agency, and shall be in accordance with paragraph 4.2.1.2 of Specification MIL-T-945. The procuring agency, however, reserves the right to make additional preproduction tests other than those outlined in Specification MIL-T-945 to insure compliance with this specification and the contract.

4.2.2 Presubmission Tests - Prior to the submission of any models for preproduction testing, the contractor shall furnish facilities for and shall conduct tests as required in paragraph 4.2.4 of Specification MIL-T-945. In addition, the data required in paragraph 4.2.3 of Specification MIL-T-945 shall be submitted to the procuring agency for review and approval prior to submission of any models.

4.2.3 Approval of Preproduction Models - Approval of preproduction model shall be by the procuring agency upon satisfactory completion of all tests. No additional models shall be delivered prior to the approval of the preproduction model. Prefabrication of any equipments prior to the approval of the preproduction model is at the contractor's own risk. One approved preproduction model will be returned to the contractor for his use in the fabrication and testing of the models to be submitted for acceptance under the contract. The preproduction model shall not be considered as one of the models under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production model.

4.3 Inspection Tests - The contractor shall furnish all models and shall be responsible for accomplishing the required tests. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the cognizant Naval Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the procuring agency. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.3.1 Scope of Inspection Tests - Inspection tests shall meet all the requirements of MIL-T-945, and shall be of the following two types:

(1) Individual Tests - Individual tests shall be conducted on each model submitted for acceptance under the contract.

(2) Sampling Tests - Sampling tests shall be conducted on a quantity as approved by the procuring agency. Models submitted for sampling tests shall be selected by the Government Inspector and shall first have passed the individual tests. Sampling tests shall be more extensive than the individual tests and may include any of the tests listed under preproduction tests which are deemed necessary by the inspector to determine that production models continue to be equivalent in performance and construction of the approved preproduction models.

4.4 Procedures - The procedures for conducting presubmission testing of preproduction models and inspection testing of production models shall be determined by the contractor. These test procedures shall be subject to review and approval by the procuring agency prior to the initiation of any testing.

4.5 Acceptance of Production Models - Acceptance of all production models shall be by the cognizant Naval Inspector.

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5. PACKAGING AND PREPARATION FOR SHIPMENT

5.1 Packaging - The equipment and spare parts shall be preserved, packaged, and packed in accordance with the requirements of Specification JAN-P-658 and shall be marked in accordance with the requirements of Specification MIL-STD-129. Suitable interior packing, blocking, and bracing shall be used to prevent shifting in the exterior shipping container and shock from handling in transit.

6. NOTES

6.1 Ordering Information - Invitation for bids and contracts or orders shall specify the title, number, and date of this specification.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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